

Exercices on session *Statistics with R - Distributions and hypothesis*

Load the package tidyverse

Admission in graduate school in UC Berkeley

We want to know if the admissions in graduate school in the Department A of UC Berkeley depends on the gender.

- The data are called *UCBAdmissions* and are available in base R. Load the data and look at their structure
- Get the data for department A (the first element of the array so you need to use the following command)

```
UCBAdmissions[, , 1]
```

- These data give the number of admitted or rejected people per gender. How is this type of table called?
- Perform a χ^2 test to see if the admission depends on the gender, and interpret it
- Check the expected frequency under the null hypothesis. Is the χ^2 test reliable?
- Get the residuals and interpret them
- Represent the residuals graphically

Weight of cabbages from different cultivars

We want to know if the weight of cabbages planted on a certain day depends on the cultivar.

- Load the dataset called *cabbages* in the package *MASS*
- Make a dataset containing only the cabbages planted at date *d21*
- Make a boxplot to graphically explore if the weight depends on the cultivar. What do you think?
- Check the normality of the distribution, using a test and graphically
- Perform a t-test
- Get the p-value, the mean estimated difference between the two groups and its confidence interval. What do you conclude?

Effect of the planting date on the content of vitamin C in cabbages

We want to know if the content of vitamin C in cabbages of the cultivar c52 depends on the planting date.

- Make a dataset containing only the cabbages of the cultivar *c52*
- Make a boxplot to explore the question graphically. What do you think?
- Perform an ANOVA
- Check the model assumptions
- Get the estimated mean of each groups
- Perform a Tukey test see which groups are different, and interpret it
- Represent the results of the ANOVA graphically

Brain and body weights in mamals

We want to know if there is a correlation between the brain and the body weights in mamals.

- Load the dataset called *mammals* in the package *MASS*
- Get the Pearson correlation coefficient
- Do a Pearson correlation test and interpret it